

IGCSE Extended Physics: Core Instructional Notes

Topic: Space Physics (Astrophysics, Orbital Mechanics, and Stellar Evolution)

1. Orbital Mechanics & Velocity Derivations

For any satellite, moon, or planet maintaining a stable, circular orbit around a massive central body, the **gravitational force of attraction** acts as the required **centripetal force** keeping it in motion.

Deriving Orbital Speed (v)

We can mathematically derive the exact speed required to maintain a stable orbit by setting the centripetal force formula equal to Newton's law of universal gravitation:

1. **Centripetal Force:**

$$F_c = \frac{m \cdot v^2}{r}$$

2. **Gravitational Force:**

$$F_g = \frac{G \cdot M \cdot m}{r^2}$$

Where:

- m = mass of the orbiting satellite or planet (kg)
- M = mass of the heavy central body being orbited (e.g., the Sun or Earth) (kg)
- v = orbital linear speed (m/s)
- r = radius of the circular orbit from the center of the mass (m)
- G = Universal Gravitational Constant ($6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$)

Setting $F_c = F_g$ balances the system parameters:

$$\frac{m \cdot v^2}{r} = \frac{G \cdot M \cdot m}{r^2}$$

Cancel out the satellite's small mass (m) and one factor of the orbital radius (r) from both sides:

$$v^2 = \frac{G \cdot M}{r}$$

Taking the square root yields our definitive orbital speed equation:

$$v = \sqrt{\frac{G \cdot M}{r}}$$

> **Exam Insight:** Notice that the required orbital speed depends **only** on the mass of the heavy central body (M) and the orbital radius (r). The mass of the orbiting satellite (m) completely cancels out and has zero effect on the velocity needed to sustain the orbit.

2. Stellar Telemetry & The Lifecycle of Stars

All stars form inside massive interstellar clouds of gas and dust called **nebulas**, pulled together by the relentless force of gravity. The path a star takes through its lifecycle is determined entirely by its initial **mass**.

Path A: Low-Mass Stars (Similar to our Sun)

1. **Protostar:** Gravitational collapse heats up the dense gas core.
2. **Stable Main Sequence Star:** Longest phase. Inward gravitational forces are perfectly balanced by outward thermal radiation pressure generated by stable **hydrogen-to-helium nuclear fusion** in the core.
3. **Red Giant:** Core hydrogen runs out. The core collapses under gravity and heats up, causing the outer shells of hydrogen to expand and cool, shifting color to red.
4. **Planetary Nebula:** The outer gas layers are ejected into space as the fusion engines destabilize.
5. **White Dwarf:** The hot, extremely dense remaining core cools down slowly over billions of years with no further fusion occurring.

Path B: High-Mass Stars (Massive Stars)

1. **Protostar $\rightarrow$ Massive Main Sequence Star:** Fuses hydrogen at an incredibly rapid rate due to extreme core pressures.
2. **Red Supergiant:** Hydrogen is depleted quickly. The star undergoes successive fusion stages, creating heavier elements up to Iron (Fe).
3. **Supernova:** Once the core turns to iron, it can no longer generate outward nuclear fusion energy. The star collapses catastrophically under its own immense gravity in seconds, triggering a massive shockwave explosion that releases heavier elements into interstellar space.
4. **The Remnants:**
 - **Neutron Star:** If the remaining core mass is moderate, electrons and protons are crushed into pure neutrons, creating an ultra-dense stellar corpse.
 - **Black Hole:** If the remaining core mass is exceptionally high (greater than approx. 3 solar masses), gravitational collapse completely overcomes neutron degeneracy pressure. The core collapses into an infinitely dense gravitational singularity from which not even light can escape.

3. Step-by-Step Mathematical Worked Example

Question: A research satellite orbits a distant planet at a steady orbital radius (r) of $8.5 \times 10^6 \text{ m}$ from the planet's core center. If the total mass of the central planet (M) is known to be $4.8 \times 10^{24} \text{ kg}$, calculate the satellite's stable orbital velocity.

Solution:

1. State the required orbital velocity formula:

$$v = \sqrt{\frac{G \cdot M}{r}}$$

2. Identify your given parameters:

- $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$
- $M = 4.8 \times 10^{24} \text{ kg}$
- $r = 8.5 \times 10^6 \text{ m}$

3. Substitute the metrics into the formula framework:

$$v = \sqrt{\frac{(6.67 \times 10^{-11}) \times (4.8 \times 10^{24})}{8.5 \times 10^6}}$$

4. Calculate the intermediate value under the radical:

$$\text{Numerator} = 3.2016 \times 10^{14}$$

$$\text{Fractional Division} = \frac{3.2016 \times 10^{14}}{8.5 \times 10^6} \approx 3.7666 \times 10^7$$

5. Compute the final square root evaluation:

$$v = \sqrt{3.7666 \times 10^7} \approx 6137.26 \text{ m/s}$$